

Practical Wise Lab Assignment x MLTL_CM23023/TensorFlows x Linear Regression HTML Code x Predict Unseen Values x +

File D:/MLTL_CM23023/Linear%20Regression%20with%20Synthetic%20Data/Lab%20Assignment%203/part3_predict_unseen.html

Prediction for Unseen Inputs

Expected output $\approx y = 2x + 1$

Elements

Sources

Network

Console

```
> predict()
< Promise {<pending>}
Prediction completed part3_predict_unseen.html:34
> unseenX.print()
Tensor tfjs@4.10.0:17
[[6 ],
 [7 ],
 [10]]
< undefined
> predictions.print()
Tensor tfjs@4.10.0:17
[[12.9896069],
 [14.9852571],
 [20.9722099]]
< undefined
> 2*6 + 1
2*7 + 1
2*10 + 1
< 21
> |
```

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Practical Wise Lab Assignment x MLTL_CM23023/TensorFlows x Learning Rate Analysis x Linear Regression HTML Code x +

File D:/MLTL_CM23023/Linear%20Regression%20with%20Synthetic%20Data/Lab%20Assignment%202/part2_learning_rate_analysis.html

Learning Rate Experiment

LR = 0.01 | |

Check Console for loss values

Elements

Sources

Network

Console

```
> testLR(1)
Training with learning rate: 1 part2_learning_rate_analysis.html:28
< Promise {<pending>}
Epoch 0 | Loss: 57.42896270751953 part2_learning_rate_analysis.html:34
Epoch 1 | Loss: 29477.466796875 part2_learning_rate_analysis.html:34
Epoch 2 | Loss: 15138498 part2_learning_rate_analysis.html:34
Epoch 3 | Loss: 7774557184 part2_learning_rate_analysis.html:34
Epoch 4 | Loss: 3992716574720 part2_learning_rate_analysis.html:34
Epoch 5 | Loss: 2050506776117248 part2_learning_rate_analysis.html:34
Epoch 6 | Loss: 1053061865707405300 part2_learning_rate_analysis.html:34
Epoch 7 | Loss: 548812579821716500000 part2_learning_rate_analysis.html:34
Epoch 8 | Loss: 2.777486459428359e+23 part2_learning_rate_analysis.html:34
Epoch 9 | Loss: 1.4263704124985749e+26 part2_learning_rate_analysis.html:34
Epoch 10 | Loss: 7.325296637058882e+28 part2_learning_rate_analysis.html:34
Epoch 11 | Loss: 3.761992608560653e+31 part2_learning_rate_analysis.html:34
Epoch 12 | Loss: 1.9320160849055184e+34 part2_learning_rate_analysis.html:34
Epoch 13 | Loss: 9.922180269130587e+36 part2_learning_rate_analysis.html:34
Epoch 14 | Loss: Infinity part2_learning_rate_analysis.html:34
Epoch 15 | Loss: Infinity part2_learning_rate_analysis.html:34
Epoch 16 | Loss: Infinity part2_learning_rate_analysis.html:34
Epoch 17 | Loss: Infinity part2_learning_rate_analysis.html:34
Epoch 18 | Loss: Infinity part2_learning_rate_analysis.html:34
Epoch 19 | Loss: Infinity part2_learning_rate_analysis.html:34
Epoch 20 | Loss: Infinity part2_learning_rate_analysis.html:34
Epoch 21 | Loss: Infinity part2_learning_rate_analysis.html:34
Epoch 22 | Loss: Infinity part2_learning_rate_analysis.html:34
```

Learning Rate Experiment

LR = 0.01

LR = 0.1

LR = 1.0

Check Console for loss values

```
> testLR(0.1)
Training with learning rate: 0.1
Epoch 0 | Loss: 95.66980743488203
Epoch 1 | Loss: 178.5552673339844
Epoch 2 | Loss: 333.2684675292969
Epoch 3 | Loss: 622.814484296875
Epoch 4 | Loss: 1160.9671638859375
Epoch 5 | Loss: 2166.909912109375
Epoch 6 | Loss: 4044.479248846875
Epoch 7 | Loss: 7548.9189453125
Epoch 8 | Loss: 14089.8720783125
Epoch 9 | Loss: 26298.40625
Epoch 10 | Loss: 49805.33984375
Epoch 11 | Loss: 91616.6171875
Epoch 12 | Loss: 171000.234375
Epoch 13 | Loss: 319167.78125
Epoch 14 | Loss: 595719.125
Epoch 15 | Loss: 1111895.625
Epoch 16 | Loss: 2075327.25
Epoch 17 | Loss: 3873548.5
Epoch 18 | Loss: 7229886.5
Epoch 19 | Loss: 13494415
Epoch 20 | Loss: 25187906
Epoch 21 | Loss: 47018956
Epoch 22 | Loss: 87744856
Epoch 23 | Loss: 163773712
Epoch 24 | Loss: 305679840
Epoch 25 | Loss: 570544064
Epoch 26 | Loss: 1064907072
Epoch 27 | Loss: 1987633800
```

Actual vs Predicted Values

Train & Plot

Open Console (F12) to see output

```
> run()
Training done
> ys.print()
Tensor
[[ 3 ],
 [ 5 ],
 [ 7 ],
 [ 9 ],
 [11],
 [13],
 [15],
 [17]]
> preds.print()
Tensor
[[2.7196689 ],
 [4.7883159 ],
 [6.8409634 ],
 [8.9015104 ],
 [10.962574 ],
 [13.0229053],
 [15.0839524],
 [17.1441994]]
```

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D:/MLTL_CM23023/Linear%20Regression%20with%20Synthetic%20Data/Lab%20Assignment%201/part1_plot_actual_vs_predicted.html

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